

Math 110 – Homework 7 Answers

Math 110 – Fall 2026

Homework 7

Diagonalizability and the minimal polynomial

Suggested completion date: Friday, October 16

Unless otherwise indicated, all vector spaces are over the field $\mathbb{F}$. Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

Problem 1. Return to the matrices from Homework 6

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

- (a) Find the minimal polynomial of A over $\mathbb{R}$ and over $\mathbb{C}$.
- (b) Find the minimal polynomial of B over $\mathbb{R}$.
- (c) Use the minimal-polynomial criterion for diagonalizability to recover the conclusions of Homework 6 about whether these operators are diagonalizable over the indicated fields.

Problem 2. Let $D : \mathcal{P}_3(\mathbb{R}) \rightarrow \mathcal{P}_3(\mathbb{R})$ be differentiation.

- (a) Find the minimal polynomial of D .
- (b) Decide whether D is diagonalizable.

Problem 3. Let T be an invertible operator on a finite-dimensional vector space, and let

$$m_T(x) = x^d + a_{d-1}x^{d-1} + \cdots + a_1x + a_0$$

be its minimal polynomial.

- (a) Prove that $a_0 \neq 0$.
- (b) Prove that T^{-1} is a polynomial in T .
- (c) Suppose now that V is a complex vector space and that

$$m_T(x) = x^4 + x^3 - x^2 - 5x + 4.$$

Express T^{-1} explicitly as a polynomial in T , find the eigenvalues of T , and decide whether T is diagonalizable.

Problem 4 (optional). Let m_T be the minimal polynomial of an operator T .

- (a) Prove that the minimal polynomial of $T - \lambda I$ is $m_T(x + \lambda)$.
- (b) If $c \neq 0$ and $d = \deg m_T$, prove that the minimal polynomial of cT is

$$c^d m_T(x/c).$$

Problem 5. Let A be a 2×2 complex matrix whose only eigenvalue is λ . Prove that its minimal polynomial is either

$$x - \lambda \quad \text{or} \quad (x - \lambda)^2.$$

Show that the first case occurs exactly when $A = \lambda I$, and conclude that A is diagonalizable if and only if $A = \lambda I$.

Problem 6. Consider

$$C = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Find its eigenvalues, eigenspaces, and minimal polynomial. Decide whether C is diagonalizable.

Problem 7 (optional). Let T be an operator whose minimal polynomial splits as

$$m_T(x) = (x - \lambda_1) \cdots (x - \lambda_r)$$

with distinct roots. For a fixed j , construct a polynomial p_j such that $p_j(\lambda_j) = 1$ and $p_j(\lambda_k) = 0$ for $k \neq j$, and explain why $p_j(T)$ is the projection onto $E(\lambda_j, T)$ along the sum of the other eigenspaces.

Solutions

Problem 1

Solution notes

Problem 2

Solution notes

Problem 3

Solution notes

Problem 4

Solution notes

Problem 5**Solution notes**

Problem 6**Solution notes**

Problem 7**Solution notes**
